

Chapter 19: Duality in Convex Optimization

Convex Analysis and Monotone Operator Theory

Based on Bauschke & Combettes (2017)

- **19.1: Primal Solutions via Dual Solutions**
 - Using dual solutions to recover primal solutions
 - Conditions for unique primal solution
 - Fenchel-Rockafellar duality framework
- **19.2: Parametric Duality**
 - Abstract dual framework
 - Value functions and weak duality
 - Strong duality conditions
- **19.3: Lagrangian Saddle Points**
 - Lagrangian function and saddle points
 - Optimality conditions
 - Connection to Lagrange multipliers

The Primal and Dual Problems

Primal Problem

Minimize $f(x) + g(Lx)$ where $f \in \Gamma_0(\mathcal{H})$, $g \in \Gamma_0(\mathcal{K})$, $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$

Dual Problem

Minimize $f^*(\bullet) + g^*(\bullet)$ over appropriate dual variables

Key Questions:

- ① How are primal and dual solutions related?
- ② Can we recover the primal solution from the dual solution?
- ③ When is the duality gap zero?

Theorem 19.1: Primal-Dual Solutions

Main Result

Let $f \in \Gamma_0(\mathcal{H})$, $g \in \Gamma_0(\mathcal{K})$, $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$. Let $x \in \mathcal{H}$, $v \in \mathcal{K}$. These are equivalent:

- ① x is a primal solution, v is a dual solution, and $\mu = -\mu^*$
- ② $-L^*v \in \partial f(x)$ and $v \in \partial g(Lx)$
- ③ $x \in \partial f^*(-L^*v) \cap L^{-1}(\partial g^*(v))$

Insight: Optimality is characterized by subdifferential conditions linking the primal and dual variables.

Proposition 19.4 & 19.5: Unique Solutions

Uniqueness via Differentiability

If f^* is Gâteaux differentiable at $-L^*v^*$ (where v^* solves the dual), then the primal problem has either no solution or a **unique** solution:

$$x^* = \nabla f^*(-L^*v^*)$$

Projection Formula

When strongly convex regularization is added to the primal:

$$\text{minimize}_x \varphi(x) + \psi(Lx - r) + \frac{1}{2}\|x - z\|^2$$

If v^* solves the dual, then:

$$x^* = \text{Prox}_{\varphi}(z - L^*v^*)$$

Example: Least Norm Problem

```
import numpy as np
from scipy.optimize import minimize_scalar

# Problem: min (1/2)||x-z||^2 s.t. <a/x> = b

z = np.array([3.0, 1.0])
a = np.array([1.0, 1.0])
b = 2.0

# Closed-form dual solution
nu_star = (z @ a - b) / (a @ a)

# Recover primal solution
x_star = z - nu_star * a
mu_primal = 0.5 * np.sum((x_star - z)**2)

# Verify duality gap is zero
h_nu = lambda nu: 0.5 * (a @ a) * nu**2 + nu * (b - z @ a)
mu_dual = h_nu(nu_star)

print(f"Primal: x* = {x_star}, mu = {mu_primal}")
print(f"Dual: nu* = {nu_star}, mu* = {mu_dual}")
print(f"Gap: {mu_primal + mu_dual}")
```

Definition 19.11: Abstract Duality Framework

General Setup

Given $F : \mathcal{H} \times \mathcal{K} \rightarrow [-\infty, +\infty]$:

- **Primal:** $\min_x F(x, 0)$
- **Dual:** $\min_v F^*(0, v)$
- **Value function:** $\vartheta(y) = \inf_x F(x, y)$

where $F^*(u, v) = \sup_{x,y} (\langle x|u\rangle + \langle y|v\rangle - F(x, y))$

Weak Duality (Proposition 19.12)

The Duality Inequality

For any convex optimization problem:

$$\inf_x F(x, 0) \geq - \min_v F^*(0, v)$$

The gap between these values is the **duality gap**.

Key Points:

- 1 Weak duality always holds for convex problems
- 2 The dual provides a lower bound on the primal optimum
- 3 The duality gap can be positive (weak duality)
- 4 Or zero (strong duality)

Strong Duality (Proposition 19.13)

Strong Duality Condition

Suppose $F \in \Gamma_0(\mathcal{H} \oplus \mathcal{K})$ and the value function ϑ is lower semicontinuous at 0 with $\vartheta(0) \in \mathbb{R}$.
Then:

$$\inf F(\mathcal{H}, 0) = - \inf F^*(0, \mathcal{K}) \in \mathbb{R}$$

Sufficient Conditions:

- $0 \in \text{int}(\text{dom } \vartheta)$ (interior)
- $0 \in \text{core}(\text{dom } \vartheta)$ (core)
- $0 \in \text{sri}(\text{dom } \vartheta)$ (strong relative interior)

Propositions 19.14 & 19.15: Dual Solutions

Characterization of Dual Solutions

The set of dual solutions is: $U = \partial\vartheta^*(0)$

Under strong duality: $U = \partial\vartheta(0)$ and $\inf F(\mathcal{H}, 0) = -\min F^*(0, \mathcal{K})$

Optimality Conditions

For $(x, v) \in \mathcal{H} \times \mathcal{K}$, the following are equivalent:

- ① x is a primal solution, v is a dual solution, strong duality holds
- ② $F(x, 0) + F^*(0, v) = 0$
- ③ $(0, v) \in \partial F(x, 0)$
- ④ $(x, 0) \in \partial F^*(0, v)$

Definition 19.16: Lagrangian

The Lagrangian Function

$$\mathcal{L}(x, v) = \inf_y (F(x, y) - \langle y | v \rangle)$$

A saddle point (x^*, v^*) satisfies:

$$\sup_w \mathcal{L}(x^*, w) = \mathcal{L}(x^*, v^*) = \inf_z \mathcal{L}(z, v^*)$$

Interpretation:

- Concave in v for each x
- Convex in x for each v (when F is convex)
- Saddle point encodes both primal and dual optimality

Propositions 19.17-19.19: Saddle Points

Connection to Dual Problems

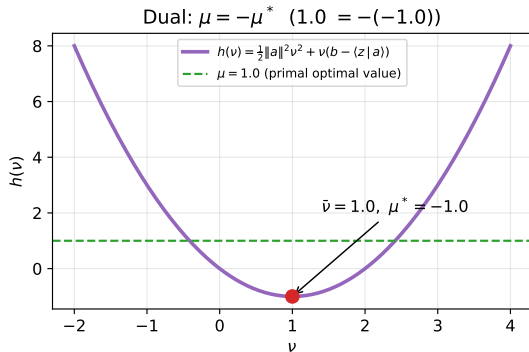
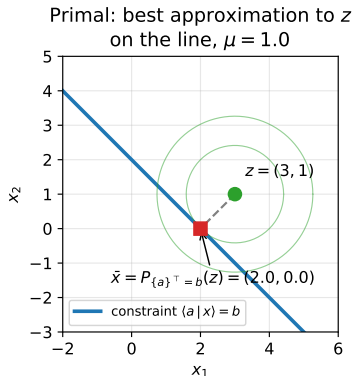
- ① For convex F : $\sup_v \mathcal{L}(x, \mathcal{K}) = F(x, 0)$
- ② For any $v \in \mathcal{K}$: $\inf \mathcal{L}(\mathcal{H}, v) = -F^*(0, v)$
- ③ Dual solution is characterized by: $\inf_w \mathcal{L}(\mathcal{H}, w) = \sup_w \inf_z \mathcal{L}(z, w)$

Complete Optimality Conditions

(x^*, v^*) is optimal if and only if (x^*, v^*) is a saddle point of $\mathcal{L}$.

Figures: Equality-Constrained Example

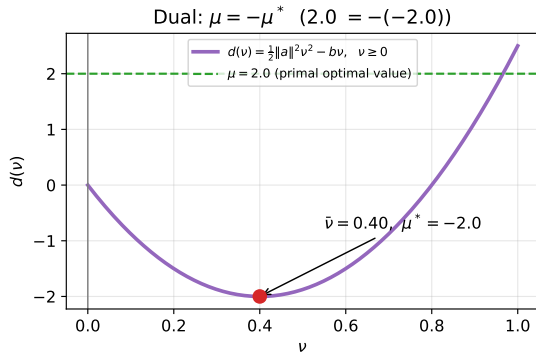
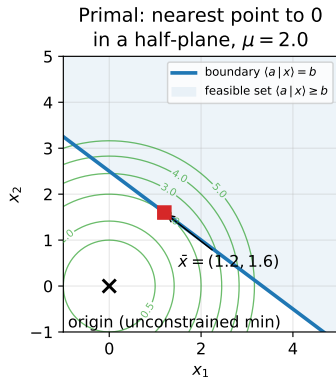
Running example (Cor. 19.23): equality-constrained least-norm problem and its dual



Running example (Corollary 19.23): Projection onto a line. Left: Primal geometry; Right: Dual objective with zero duality gap.

Figures: Inequality-Constrained Example

Running example (Cor. 19.30): inequality-constrained problem and its dual



Running example (Corollary 19.30): Projection onto a half-plane. Left: Feasible region and optimal solution; Right: Dual objective.

Why Duality Matters

Computational Benefits

- 1 **Tractability:** Dual often easier than primal
- 2 **Smoothness:** Dual objectives are differentiable (Moreau envelopes)
- 3 **Decomposition:** Enables distributed/parallel algorithms (ADMM, proximal methods)
- 4 **Bounds:** Dual provides lower bounds for verification

Key Applications

- Support vector machines (SVM)
- Semidefinite programming
- Distributed optimization

Key Takeaways

- 19.1** Primal solutions are uniquely recovered from dual solutions via the Fenchel-Rockafellar framework
- 19.2** Weak duality always holds; strong duality holds under regularity conditions (lower semicontinuity)
- 19.3** Lagrangian saddle points characterize optimal primal-dual pairs and connect to Lagrange multiplier theory

Practical Impact: Duality transforms difficult primal problems into more tractable dual formulations, enabling efficient algorithms for large-scale optimization.

Next: Monotone operators (Chapter 20) provide the operator-theoretic foundation for algorithm development.